

Data Science Practical

Case 1 - Academic year 2025-2026

Explaining the salaries of CEOs

The compensation for a lot of jobs is determined through collective labor agreements. In most cases, this means that no salary adjustments are introduced even when the employee or the firm is doing particularly well. For Chief Executive Officers (CEOs), salaries are typically much more variable. The job requires a very specific skill set which often translates into a limited supply of CEOs. For this reason, they have a lot of bargaining power over the firms that pay their salary. Another important consideration in this situation is the so-called 'agency problem': the CEO is part of the company's management which is supposed to act in the best interest of the company's stakeholders. In order to tackle this problem, it is often the case that the CEOs compensation is determined by a measure or several measures of firm's performance. You hypothesize that return on equity is a proper measure.

Estimating a simple linear regression could be a nice starting point, but is probably not very realistic as the CEO's compensation is modeled in such a way that it only depends on a single variable. Luckily, you have more information available in your data set. Before you continue your analysis, you notice that salary and sales are available in normal levels and in log-levels (why?). After that, you start thinking about what variables might serve as explanatory variables (for the moment, disregard the dummy variables at your disposal) and build a new model including them. You want to assess the new model's adequacy to explain the CEO's compensation and you want to show that it provides a better fit than the simple linear regression model. To that end, you employ a couple of measures and tests. Additionally, you ask yourself whether two (or more) measures of firm's performance have the same effect on the compensation of the CEO. In every step of the analysis, you make sure that you explain everything in solid statistical and economic terms, since you are planning on sharing your results with the rest of the world. Hence, your story should convince experts and be clear to laymen. A careful explanation of (the interpretation of) estimation results is important for both.

Despite the fact that your newly built model probably "outperforms" the previous one, you are still not satisfied. You remember that the linear regression model is a pow-

erful tool that can also deal with non-linearity and qualitative variables. For this reason, you use the model you built last and check for any empirical signs for non-linear relationships. Concerning the qualitative variables, you critically decide on which ones you need (avoid a dummy trap!) and add them to the model. At the same time, you think of something else. Maybe the compensation of the CEO does not depend on the return on the firm's stock directly, but maybe it only matters whether the return was negative or not?

Lastly, you realize that the data set you studied was definitely useful, but might also have clear limitations. Consequently, you start thinking about other factors that could explain the compensation of the CEO. Since you want to make sure that the results of your study are reliable, you believe that it might be appropriate to check for some common problems that may occur in empirical studies (are the assumptions underlying the regression model met?).